Internship Report

ElevanceSkills — Generative AI Internship

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Role: Generative AI Intern
Duration: 1 Month

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1. Introduction

This report summarizes my learning journey, tasks, challenges, and achievements during my one-month Generative AI internship at ElevanceSkills. The internship provided practical exposure to modern AI technologies, especially in the areas of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, multimodal AI systems, and domain-specific chatbot development.

During this internship, I enhanced my previously developed AI Customer Service Bot by implementing multiple advanced AI capabilities. The final result was a Multi-Mode AI Assistant capable of handling customer support, medical question answering, research paper analysis, multilingual conversations, and sentiment-aware interactions.

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2. Background

Artificial Intelligence is rapidly transforming how humans interact with software systems. Traditional chatbots often struggle with context retention, emotional understanding, and domain-specific reasoning.

Generative AI models such as Gemini, combined with retrieval systems like FAISS and ChromaDB, enable the development of intelligent assistants capable of reasoning over structured and unstructured data.

Before this internship, I had foundational knowledge of Python and web development. Through the training and internship, I learned how to integrate advanced AI systems into practical applications.

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3. Learning Objectives

The major objectives of this internship were:

- Understand the architecture of LLM-powered applications
- Learn Retrieval-Augmented Generation (RAG)
- Work with vector databases such as FAISS and ChromaDB
- Develop multimodal AI systems using text and images
- Build domain-specific AI assistants
- Implement sentiment-aware conversational systems
- Add multilingual capabilities to AI chatbots
- Improve software engineering and debugging skills

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4. Activities and Tasks

During the internship, I completed six major tasks by extending my training project.

Task 1 — Dynamic Knowledge Base Expansion

I implemented a mechanism to dynamically update the chatbot knowledge base whenever source documents changed.

Implementation:

- Used ChromaDB for vector storage
- Added knowledge base change detection
- Rebuilt embeddings automatically when updates occurred

Outcome:

The chatbot can now continuously incorporate new knowledge without retraining the full system.

Task 2 — Multi-Modal AI Assistant

I enhanced the chatbot to understand both text and image inputs.

Implementation:

- Added screenshot upload feature using Streamlit
- Integrated image analysis pipeline

- Used Gemini for visual reasoning

Outcome:

The assistant can analyze screenshots and provide context-aware responses.

Task 3 — Medical Q&A Chatbot

I built a specialized medical assistant using the MedQuAD dataset.

Implementation:

- Parsed XML medical dataset
- Created FAISS vector database
- Implemented medical retrieval pipeline

Outcome:

Users can ask symptom, disease, and treatment-related questions.

Task 4 — Research Expert Chatbot

I developed a research assistant trained on AI-related arXiv papers.

Implementation:

- Processed arXiv dataset
- Filtered AI-related papers
- Built FAISS-based scientific paper retrieval system
- Added concept explanation and summarization

Outcome:

The assistant can explain advanced research topics and summarize papers.

Task 5 — Sentiment Analysis

I integrated sentiment analysis into customer conversations.

Implementation:

- Classified messages as Positive, Negative, or Neutral
- Adjusted chatbot responses based on emotional tone

Outcome:

The chatbot responds more naturally and empathetically.

Task 6 — Multilingual Support

I extended the chatbot to support multiple languages.

Supported Languages:

- English
- Hindi
- Gujarati
- Urdu

Implementation:

- Language detection
- Translation pipeline
- Cross-language context retention

Outcome:

Users can communicate in multiple languages seamlessly.

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5. Skills and Competencies

During this internship, I improved the following technical skills:

Programming

- Python
- Streamlit
- API integration

AI / Machine Learning

- Generative AI

- RAG pipelines
- Prompt engineering
- Vector embeddings
- Semantic search

Tools & Libraries

- Gemini API
- LangChain
- FAISS
- ChromaDB
- HuggingFace
- Sentence Transformers

Soft Skills

- Problem solving
- Debugging
- Technical documentation
- Research skills

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6. Feedback and Evidence

The mentorship and training sessions helped me understand industry-level AI development practices.

Evidence of work completed includes:

- GitHub repository with source code
- Test scripts for each module
- Streamlit application
- Dataset preprocessing scripts
- Vector database pipelines
- Successful execution screenshots and outputs

The project demonstrates practical application of all assigned internship tasks.

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7. Challenges and Solutions

During development, I faced multiple challenges.

Challenge 1: Large Dataset Processing

The arXiv dataset size was approximately 5GB, making preprocessing slow.

Solution:

I created a smaller AI-focused subset to improve efficiency.

Challenge 2: Vector Database Issues

FAISS installation and vector generation initially failed.

Solution:

Installed compatible FAISS version and rebuilt vector stores.

Challenge 3: Poor Retrieval Accuracy

Research queries initially returned irrelevant papers.

Solution:

Implemented query expansion and improved retrieval logic.

Challenge 4: API Rate Limits

Gemini occasionally returned quota or busy errors.

Solution:

Added model fallback and graceful error handling.

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8. Outcomes and Impact

At the end of the internship, I successfully transformed a basic chatbot into a powerful Multi-Mode AI Assistant.

Final capabilities include:

- Customer service support
- Medical question answering
- Research assistance
- Multimodal reasoning
- Sentiment-aware conversations
- Multilingual communication

This project significantly improved my understanding of practical Generative AI system design.

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9. Conclusion

The Generative AI internship at ElevanceSkills was highly valuable for my technical growth. It provided practical experience in designing intelligent AI systems using modern tools and frameworks.

Through this internship, I strengthened both my AI engineering and software development skills. The project gave me hands-on experience with real-world challenges such as data preprocessing, retrieval optimization, model limitations, and scalable architecture design.

This internship has increased my confidence in building advanced AI applications and motivated me to continue exploring the field of Generative AI and Machine Learning.